

**EXPLAINABLE MULTI-MODAL AI SYSTEM FOR EARLY
ACADEMIC BURNOUT AND DROPOUT RISK DETECTION**

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Academic burnout and dropout develop gradually and appear through several weakly connected signals, so universities usually notice them only when grades fall, after the best time to intervene has passed. This report presents R26-IT-059, an explainable multi-modal AI system that gives student advisors earlier and reason-based warnings. Four specialist components each analyse one signal: a Unified Multi-Horizon GRU predicts academic dropout risk at Weeks 4, 8, 12 and 17 from LMS activity; a BERT+RoBERTa ensemble classifies student text into seven emotional states; a Variational Autoencoder detects anomalies in LMS behaviour against a shared not-at-risk baseline; and a Meta-Integration layer fuses these signals with a feedforward network into one HIGH, MEDIUM or LOW risk score, while a Meta-XAI decision layer turns each student's risk trajectory into an explainable intervention priority and advisor action. Every component explains its output, and the fused result is served through a Flask REST API to an advisor dashboard. The academic component reached F1 0.7227 and AUC 0.8831 at Week 17, the text component reached 82.32% accuracy (weighted F1 0.8232), and the calendar-aware behavioural detector reached AUC 0.6039 on OULAD, staying above random up to eight weeks ahead. A fusion prototype reached AUC 0.8576 as a feasibility check: because the components' outputs could not yet be linked student by student, the behavioural inputs were simulated and the emotional scores were not matched to the same students, so this result shows that the pipeline works end to end rather than validated detection performance. The results show that each signal adds information and that combining them with per-student explanations can support earlier, human-reviewed intervention, pending validation on real linked student data.

Keywords: academic burnout, dropout prediction, multi-modal fusion, explainable AI, learning analytics

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TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT	1
1.1 Background and Context.....	1
1.2 Stakeholders and User Needs.....	1
1.3 Alignment with the Sustainable Development Goals	2
1.4 Purpose and Scope of the Common Report	2
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS	3
2.1 Overall Problem Definition.....	3
2.2 Root Causes and Overall Research Gap.....	3
2.3 Subproblem and Component Mapping	3
2.4 Overall Aim and Group Objectives.....	4
2.5 Scope Assumptions and Constraints	5
3 OVERALL SOLUTION STRUCTURE.....	6
3.1 Integrated Solution Overview	6
3.2 High Level Architecture.....	6
3.3 Component 1 Summary.....	7
3.4 Component 2 Summary.....	7
3.5 Component 3 Summary.....	8
3.6 Component 4 Summary.....	9
3.7 End to End User and Data Flow.....	10
4 COMPONENT INTEGRATION AND COHESIVE OPERATION.....	12
4.1 Integration Strategy	12
4.2 Interfaces Data Contracts and Shared Models	12
4.3 Shared Nonfunctional Requirements	13
4.4 Integrated Workflow and Exception Handling	13

4.5 Team Contribution and Dependency Matrix.....	14
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION	15
5.1 Common Technology Stack and Infrastructure	15
5.2 Development and Integration Process.....	15
5.3 Deployment and Operational Configuration.....	15
5.4 Evidence of Team Execution	16
6 INTEGRATED TESTING EVALUATION AND VALIDATION	17
6.1 Evaluation Framework and Success Criteria	17
6.2 Integration and End to End Testing	17
6.3 Overall Results	18
6.4 Stakeholder and User Validation	19
6.5 Achievement of Common Objectives	19
6.6 Integrated Limitations and Threats	20
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION	21
7.1 Value Proposition and Beneficiaries.....	21
7.2 Market Alternatives and Differentiation	21
7.3 Business Model and Feasibility	22
7.4 Adoption Intellectual Property and Compliance.....	22
7.5 Sustainable Development Goal Contribution.....	22
8 CONCLUSION AND FUTURE DIRECTIONS	23
8.1 Overall Conclusion.....	23
8.2 Principal Integrated Contribution.....	23
8.3 Future Directions.....	23
REFERENCES.....	24
APPENDICES.....	25

Appendix A Integrated Interface Specification.....	25
Appendix B Team Contribution Declaration	25

LIST OF FIGURES

Figure 3.1: High-level architecture of the R26-IT-059 integrated system.....	7
Figure 3.2: Data flow between a specialist component, the API, the dashboard and the meta-layer.....	11

LIST OF TABLES

Table 2.1: Subproblem and component mapping	3
Table 2.2: Group objectives, components and evaluation evidence	4
Table 3.1: Academic Trend Prediction component summary	7
Table 3.2: Emotional Burnout Detection component summary	7
Table 3.3: Behavioural Anomaly Detection component summary	8
Table 3.4: Meta-Integration and Meta-XAI Decision Layer component summary	9
Table 4.1: Shared data contract	12
Table 4.2: Shared non-functional requirements	13
Table 4.3: Team contribution and dependency matrix	14
Table 5.1: Common technology stack	15
Table 6.1: Integration and end-to-end tests	17
Table 6.2: Summary of component and integrated results	18
Table 6.3: Achievement of group objectives	19
Table 7.1: Comparison with existing solutions	21
Table A.1: REST API endpoints	25
Table B.1: Team contribution declaration	25

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the ROC Curve
BERT	Bidirectional Encoder Representations from Transformers
CSV	Comma-Separated Values
FNN	Feedforward Neural Network
GRU	Gated Recurrent Unit
JSON	JavaScript Object Notation
LMS	Learning Management System
NLP	Natural Language Processing
OULAD	Open University Learning Analytics Dataset
PDPA	Personal Data Protection Act
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
VAE	Variational Autoencoder
VLE	Virtual Learning Environment
XAI	Explainable Artificial Intelligence

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Student dropout is a persistent problem in higher education, and academic burnout, a state of exhaustion, cynicism and reduced efficacy caused by prolonged study pressure, is one of its main drivers [1]. Online and blended learning has increased the volume of data universities hold about students, including learning management system (LMS) activity, assessment results and written communication [2]. However, this data is rarely combined. Advisors usually learn that a student is struggling only after grades have fallen or the student has stopped attending, when the most effective time to intervene has already passed.

Burnout and dropout risk appear across several weakly connected signals: declining academic engagement, emotional distress in what students write, and changes in online behaviour. Each signal on its own is incomplete. The R26-IT-059 project therefore combines these signals in one explainable system that warns advisors earlier and tells them why a student was flagged.

1.2 Stakeholders and User Needs

The primary users are academic and student-support advisors, who decide which students to contact and what support to offer. Secondary stakeholders are students, who benefit from earlier support; university management, who are concerned with retention; and counselling services, who receive escalated cases. The needs identified for these stakeholders are listed below.

- **Early warning:** a list of at-risk students while there is still time to act.
- **Prioritisation:** a clear HIGH, MEDIUM or LOW level so that limited advisor time goes to the most urgent cases.
- **Explanation:** the reason behind each alert, such as falling logins or distress in written text, so the response can be targeted.
- **Human control:** the system must support, not replace, the advisor's judgement, and sensitive cases must be escalated to qualified staff.
- **Privacy:** student data must be anonymised and access restricted.

A formal stakeholder survey was not conducted; these needs were derived from the literature and refined through supervisor feedback at the progress presentations.

1.3 Alignment with the Sustainable Development Goals

The project contributes mainly to SDG 4 (Quality Education), particularly target 4.3 on equal access to quality tertiary education, by helping institutions identify and support students before they withdraw. Through its emotional-distress component it also supports SDG 3 (Good Health and Well-being), target 3.4 on mental health and well-being. These are intended contributions; measurable impact would require deployment and evaluation in a real institution (Section 7.5).

1.4 Purpose and Scope of the Common Report

This common report presents the shared problem, the integrated solution and the system-level evaluation of R26-IT-059. It explains how the four components fit together, how they are integrated and what the combined system achieves. Detailed literature reviews, methods, experiments and component-level testing are reported in the four individual theses, which this report references rather than repeats.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

Universities do not detect students at risk of academic burnout and dropout early enough, and when an alert is raised advisors do not know why the student is at risk. Existing early warning systems rely on a single data source, usually grades or LMS activity, predict at a single point in time and give little or no explanation, so the resulting interventions are late and generic.

2.2 Root Causes and Overall Research Gap

Three root causes underlie the problem:

- **Single-signal monitoring:** academic, emotional and behavioural signs are monitored separately or not at all, so students who are struggling in only one dimension are missed.
- **Late or single-point prediction:** most systems predict at one checkpoint, often near the end of a course, without showing when an alert first becomes reliable.
- **Opaque models:** most predictive models give a risk score without a reason, which advisors cannot act on or trust [10].

The overall research gap is the absence of an explainable system that combines academic, emotional and behavioural signals, predicts risk early and at several points in time, and explains each alert to advisors at the level of the individual student. Closing this gap requires four complementary components rather than a single model.

2.3 Subproblem and Component Mapping

The common problem is divided into four non-overlapping but interdependent subproblems, each addressed by one team member (Table 2.1). Each component covers a signal the others cannot observe, and the fusion layer depends on all of them; removing any one would leave a type of at-risk student undetected.

Table 2.1: Subproblem and component mapping

Subproblem	Component	Owner
When does academic dropout risk become detectable, and why is a student at risk?	Academic Trend Prediction	Mahavithana S. M. (IT22916426)
Which students show emotional distress in what they write?	Emotional Burnout Detection	Induwara K. P. Y. (IT22196392)
Which students' LMS behaviour deviates from normal engagement, without clinical labels?	Behavioural Anomaly Detection	Karunaratne D. C. (IT22215710)
How can all signals be combined into one explainable risk and turned into prioritised advisor action?	Meta-Integration and Meta-XAI Decision Layer	Jayawickrama G. T. (IT22253194)

2.4 Overall Aim and Group Objectives

Aim: to develop and validate an explainable multi-modal AI system that detects students at risk of academic burnout and dropout earlier than single-signal approaches and explains each alert to advisors.

Table 2.2: Group objectives, components and evaluation evidence

Objective	Components	Evidence
G1 Predict academic risk at several intervention windows	Academic	F1/AUC per week (Table 6.2)
G2 Detect emotional distress from text with explanations	Emotional	Accuracy, weighted F1 (Table 6.2)
G3 Detect behavioural anomalies without clinical labels	Behavioural	AUC across 0–8 week lags (Table 6.2)
G4 Fuse all signals into one calibrated risk level	Meta-integration	Fused F1/AUC, modality contribution (Table 6.2)
G5 Deliver explanations and alerts to advisors through an API and dashboard	All	Integration tests (Table 6.1)

2.5 Scope Assumptions and Constraints

- The system is a decision-support tool; designing and delivering interventions is out of scope.
- All models were trained on public datasets: OULAD [3] for academic and behavioural signals and a public mental-health text corpus [4] for the emotional signal. No SLIIT student data was used.
- The text corpus comes from social-media posts, so it differs from real student writing (domain shift).
- Predictions are produced in weekly batches rather than in real time.
- Every alert requires human review; critical emotional cases are escalated to qualified mental-health staff.
- The components were developed on separate data, so their outputs are not yet linked student by student: in the current prototype the behavioural inputs to the fusion stage are simulated and the emotional scores are not matched to the same students; only the academic-risk inputs are genuinely linked.
- The system depends on the four components sharing a common student identifier and output format (Section 4.2).

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

The solution is a single early warning system in which four specialist models each analyse one signal and a meta-integration layer combines them. Each specialist model produces a standardised weekly output: a risk score, a level and the main reasons. The meta-integration layer fuses these outputs into one burnout risk score with a HIGH, MEDIUM or LOW alert and shows which signal contributed most. Advisors see a single list of at-risk students, each with an explanation, instead of four separate tools. The value of the system is that it detects students whom any single signal would miss and explains every alert.

3.2 High Level Architecture

Figure 3.1 shows the architecture. Data sources feed four specialist models running in parallel. Their standardised outputs are passed to the meta-integration layer, whose fused result is served by a Flask REST API to the advisor dashboard.

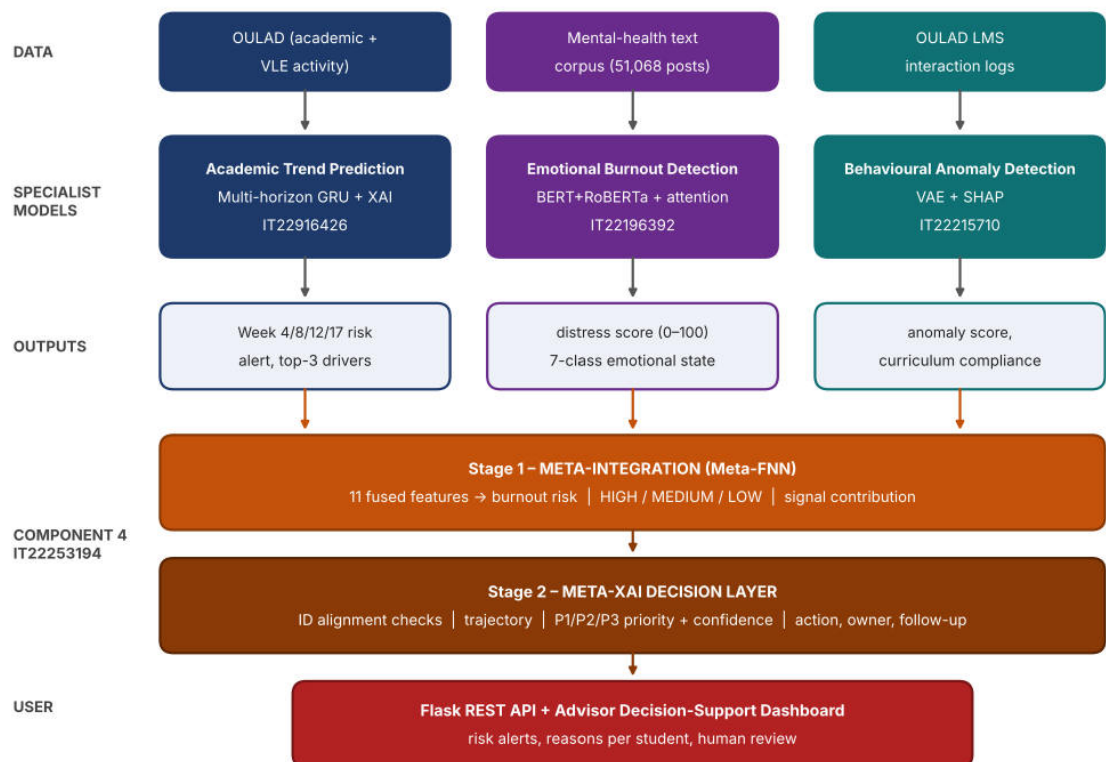


Figure 3.1: High-level architecture of the R26-IT-059 integrated system

3.3 Component 1 Summary

Academic Trend Prediction is developed by Mahavithana S. M. (IT22916426). Its role in the integrated system is summarised in Table 3.1; full details are given in the individual thesis.

Table 3.1: Academic Trend Prediction component summary

Item	Description
Subproblem	When dropout risk becomes reliably detectable and why each student is at risk.
Input	Thirteen weekly engagement and academic features from OULAD (16,950 students, 17 weeks).
Core method	Unified Multi-Horizon GRU with one shared encoder and four prediction heads (Weeks 4, 8, 12, 17); gradient and SHAP attribution for per-student explanations.
Output	Risk at four horizons, alert level and top three risk drivers per student (CSV/JSON).
Key result	Week 17 F1 0.7227, AUC 0.8831; $AUC \geq 0.75$ from Week 8; outperformed seven baselines (McNemar $p = 0.0098$).
Contribution to the system	Provides the academic-risk signal (5 of the 11 fusion inputs) and the shared Flask API and data contract.

3.4 Component 2 Summary

Emotional Burnout Detection is developed by Induwara K. P. Y. (IT22196392). Its role in the integrated system is summarised in Table 3.2; full details are given in the individual thesis.

Table 3.2: Emotional Burnout Detection component summary

Item	Description
Subproblem	Detecting emotional distress in student text with advisor-readable explanations.

Input	51,068 labelled text passages in seven classes from a public mental-health corpus [4].
Core method	Ten models compared; soft-voting ensemble of fine-tuned BERT and RoBERTa; phrase-level attention heatmaps cross-checked with LIME and SHAP on example texts.
Output	Seven-class emotional state, risk level (Low to Critical), distress score (0–100) and highlighted words for the dashboard; emotional stress score (0–4) for the fusion layer.
Key result	Accuracy 82.32% (95% CI 81.46–83.17), weighted F1 0.8232, macro F1 0.789; significantly better than BERT ($p = 0.0002$) and Emotion-DistilRoBERTa ($p < 0.001$), statistically level with RoBERTa alone ($p = 0.96$); per-class F1 from 0.96 (Normal) to 0.65 (Personality Disorder); in a simulated lead-time test, distress and normal texts were separated with 89–91% accuracy at every checkpoint (Weeks 2–12). Labels come from the source corpora, not clinicians, so outputs are prompts for human review, never a diagnosis.
Contribution to the system	Provides the emotional signal (1 of the 11 fusion inputs) and immediate escalation of critical (Suicidal) cases.

3.5 Component 3 Summary

Behavioural Anomaly Detection is developed by Karunarathne D. C. (IT22215710). Its role in the integrated system is summarised in Table 3.3; full details are given in the individual thesis. Feature contributions are explained with SHAP [5].

Table 3.3: Behavioural Anomaly Detection component summary

Item	Description
Subproblem	Detecting abnormal LMS behaviour without sensitive clinical labels.

Input	Ten weekly LMS features from OULAD (clicks, active days, content diversity, forum, resource and quiz clicks, submission latency, late submissions, curriculum compliance, click compliance); validated also on a 500-student SLIIT synthetic dataset.
Core method	Variational Autoencoder (10→64→32→6 latent, 8,502 parameters, KL annealing) trained only on baseline weeks of normally engaged students (20,727); calendar-aware score = $0.20 \times \text{reconstruction error} + 0.80 \times (1 - \text{curriculum compliance})$; SHAP explanations.
Output	Weekly anomaly score, high-anomaly flag and feature contributions.
Key result	Version 2 combined AUC 0.6039 on OULAD (Version 1: 0.5840); 0.6134 at 0-week lag, above random up to 8 weeks ahead; curriculum compliance alone 0.6053, VAE reconstruction alone 0.4261; SLIIT synthetic dataset AUC 0.7649.
Contribution to the system	Provides the behavioural signal (5 of the 11 fusion inputs) that is available continuously without extra data collection.

3.6 Component 4 Summary

Meta-Integration and Meta-XAI Decision Layer is developed by Jayawickrama G. T. (IT22253194). Its role in the integrated system is summarised in Table 3.4; full details are given in the individual thesis.

Table 3.4: Meta-Integration and Meta-XAI Decision Layer component summary

Item	Description
Subproblem	Combining the specialist signals into one explainable risk and turning it into prioritised, actionable advice for advisors.

Input	Standardised outputs of the other three components (academic horizon risks, behavioural features, emotional distress score).
Core method	Stage 1 – Meta-Integration FNN (11 inputs, four layers) with HIGH/MEDIUM/LOW mapping, modality-contribution analysis and a transparent weighted-sum alternative. Stage 2 – Meta-XAI layer: input validation and student-ID alignment
	checks, rule-based academic trajectory classification (e.g. rapidly escalating, persistent high, improving) from the Week 4–17 risks, P1/P2/P3 intervention priority with confidence that falls when evidence is missing, recommended action, owner and case workflow; an intervention-opportunity logistic model is designed but awaits outcome data for training.
Output	Fused burnout risk, alert level, signal contributions, trajectory, P1/P2/P3 priority with reason codes, recommended action and case status.
Key result	Fusion prototype F1 0.6836, AUC 0.8576, recall 0.7833 (feasibility check); Meta-XAI workflow implemented and demonstrated on demo data.
Contribution to the system	Turns the specialist signals into one prioritised, explained case for each student and records what the advisor does next.

3.7 End to End User and Data Flow

The end-to-end flow for one weekly cycle is as follows, and the interaction between the academic component, the API, the dashboard and the meta-layer is shown in Figure 3.2.

- Weekly LMS activity and any available student text are collected and anonymised.
- Each specialist model runs in batch and writes its standardised output (CSV/JSON) keyed by the shared student identifier.

-
- The diagram illustrates the 10 steps of the AWS Well-Architected Framework, organized into five categories represented by colored boxes at the top:
- Security & Compliance** (Dark Blue): Step 10 - Build multi-account.
 - Performance & Cost** (Dark Grey): Steps 2, 3, 4, 5, 6, 7, 8, 9.
 - Reliability & Availability** (Teal): Step 3 - Scale (autoscaling).
 - Operational Excellence** (Red): Step 6 - Distribute data + requests.
 - Security & Compliance** (Orange): Step 10 - Build multi-account.
- The steps are as follows:
1. Define workload & identify key, start, stop, or delete
 2. Provision for start-up
 3. Scale (autoscaling)
 4. Store for read, retrieve
 5. Scale dynamically for throughput
 6. Distribute data + requests
 7. Release for more capacity
 8. Build multi-availability
 9. Build multi-region
 10. Build multi-account

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

The team used a late-fusion, interface-first strategy. The output format of each component was agreed before integration, so that each model could be trained, validated and explained independently and then combined by the meta-integration layer without joint training, an approach established for combining heterogeneous behavioural signals [9]. This keeps the system modular: a component can be retrained or replaced without changing the others. Integration proceeded in three steps: (1) each component exported its predictions in the agreed CSV/JSON format; (2) the meta-layer consumed these files and was trained on the combined features; (3) the outputs were exposed through the Flask API and the dashboard. The main integration risk, mismatched student identifiers between data sources, was managed by agreeing one identifier format and validating it before fusion.

4.2 Interfaces Data Contracts and Shared Models

All components exchange data through files and REST endpoints. The shared contract is summarised in Table 4.1, and the full interface specification is given in Appendix A.

Table 4.1: Shared data contract

Producer	Fields passed to the meta-layer	Format
Academic (IT22916426)	academic_risk; week4_risk, week8_risk, week12_risk, week17_risk; alert_level; top three drivers	CSV/JSON, one record per student
Emotional (IT22196392)	student_id; predicted_label; confidence; risk_level; emotional_stress_score (0–4)	CSV, one record per student (emotional_stress_scores.csv)
Behavioural (IT22215710)	mean and maximum behavioural risk, LMS compliance, mean anomaly score, high-anomaly week count	CSV, one record per student-week

Meta-layer (IT22253194)	fused risk (0–1), alert level (HIGH > 0.70, MEDIUM 0.40–0.70, LOW < 0.40), modality contributions	JSON via REST API
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Validation rules: every record must carry the shared student identifier; probabilities must lie in $[0, 1]$; alert levels must use the agreed thresholds. The producer of each file is responsible for its validity, and the API returns HTTP 400 for missing parameters and HTTP 404 for unknown students.

4.3 Shared Nonfunctional Requirements

Table 4.2: Shared non-functional requirements

Requirement	Target and how it is met
Performance	API responses within 0.5 s; measured maximum 0.09 s for the full list of 2,543 students
Explainability	Every alert carries reasons from at least one component (drivers, phrases or anomalous features)
Privacy and security	Anonymised identifiers only; role-based dashboard access; no raw text shown without need
Reliability	Batch outputs are stored as files, so the dashboard keeps working if a model is being retrained
Maintainability	Modular components with documented interfaces, README files and version control
Responsible AI	Human review of every alert; immediate escalation of critical emotional cases

4.4 Integrated Workflow and Exception Handling

In normal operation all four components produce their weekly outputs, the meta-layer fuses them and the dashboard displays the result. The main failure paths and the agreed responses are:

- **Missing component output:** the meta-layer can fall back to a transparent weighted sum ($0.4 \times \text{academic} + 0.3 \times \text{behavioural} + 0.3 \times \text{emotional}$) over the available signals, and the dashboard shows which signals were used.
- **Unknown student or invalid request:** the API returns HTTP 404 or HTTP 400 with a clear message instead of failing.
- **Predictions not loaded:** the API returns HTTP 500 with a message, and the health endpoint reports the problem.
- **Critical emotional result:** the alert is escalated immediately to a qualified professional regardless of the fused score.

4.5 Team Contribution and Dependency Matrix

Table 4.3: Team contribution and dependency matrix

Member	Research and development	Integration work	Depends on
Mahavithana S. M.	Multi-horizon GRU, explainability, baselines, validation	Flask API, CSV/JSON contract, academic dashboard view	Shared ID format
Induwara K. P. Y.	Text dataset, ten-model comparison, BERT+RoBERTa ensemble, attention explanations, risk and distress scoring, statistical tests	Emotional API (port 5004), Emotional Analysis dashboard tab, emotional stress score export for fusion	Shared ID format
Karunaratne D. C.	LMS feature engineering, VAE, lag analysis, SHAP	Behavioural features for fusion	Shared ID format
Jayawickrama G. T.	Meta-FNN fusion, contribution analysis, Meta-XAI trajectory and priority logic	Input validation and ID alignment, intervention-queue dashboard	All three component outputs
All members	Joint paper (ICAC 2026), progress presentations	End-to-end testing, this report	—

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

Table 5.1: Common technology stack

Layer	Technology	Reason
Language	Python 3	Shared by all components and ML libraries
Deep learning	PyTorch; Hugging Face Transformers	GRU, VAE, FNN and BERT/RoBERTa models
Data and ML	pandas, NumPy, scikit-learn	Feature engineering, baselines and metrics
Explainability	SHAP, LIME, attention weights	Per-student and per-signal explanations
Service	Flask REST API	Lightweight, simple integration
Frontend	HTML, CSS, JavaScript dashboard	Advisor interface
Collaboration	GitHub, Microsoft Teams, Microsoft Planner	Version control, communication, task tracking

5.2 Development and Integration Process

Development followed an iterative process aligned with the project milestones: proposal (March 2026), Progress Presentation 1 (May 2026), Progress Presentation 2 (August 2026) and the final presentation (October 2026). Each member developed and validated a component independently, while the data contract was agreed early so that integration could proceed in parallel. Source code was kept under version control on GitHub, tasks were tracked in Microsoft Planner, and supervisor reviews at each milestone led to revisions, including the correction of a data-leakage issue in the cross-year validation and a serialisation defect found during API testing.

5.3 Deployment and Operational Configuration

The system is deployed as a local prototype. Each week, the specialist models are run in batch to produce their output files; the meta-layer then produces the fused risk;

and the Flask API loads all outputs at start-up and serves them to the browser-based dashboard. Configuration consists of the file locations and the port number. Because all outputs are stored as files, backing up and restoring the system means copying the output folders together with the trained model files. Production hosting, monitoring and automated scheduling are identified as future work (Section 8.3).

5.4 Evidence of Team Execution

Team execution is evidenced by the Microsoft Planner task board (tasks from proposal to final report with start and due dates), the GitHub repository history (github.com/Sha20000/R26-IT-059-Burnout-Prediction), Microsoft Teams meetings with the supervisors, the proposal and two progress presentations, and the joint paper on multi-modal fusion submitted to ICAC 2026. Each member's research diary and individual thesis provide further detail.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

The integrated system was evaluated at two levels. At component level, each model was evaluated on held-out test data with metrics suited to its task (F1 and AUC for risk prediction, accuracy and weighted F1 for text classification, AUC across lags for anomaly detection). At system level, the fused model was evaluated on a held-out split, and the integration was verified with interface and end-to-end tests. The success criteria were: $AUC \geq 0.75$ for risk prediction; a behavioural signal consistently above random; a fused AUC above that of any single non-academic signal; and all integration tests passing.

6.2 Integration and End to End Testing

Integration testing focused on the shared contract and the API (Table 6.1). Component-level unit tests are reported in the individual theses.

Table 6.1: Integration and end-to-end tests

ID	Test	Expected result	Status
IT-01	Academic CSV and JSON exports match	Same 2,543 records and values	Pass
IT-02	API health check	HTTP 200; predictions loaded	Pass
IT-03	Filter by alert level	HTTP 200; correct subset (569 HIGH)	Pass
IT-04	Single student by index or ID	HTTP 200; JSON record	Pass (after serialisation fix)
IT-05	Invalid index or unknown ID	HTTP 404 / HTTP 400	Pass
IT-06	Meta-layer reads component outputs	Fusion vector built without format errors	Pass
IT-07	Dashboard shows fused alerts and reasons	Risk cards, alert levels and drivers displayed	Pass
IT-08	Fusion with real, linked text and	All 11 inputs from real per-student outputs	Pending (no linked dataset)

	behavioural inputs		
IT-09	Emotional API health check	HTTP 200; BERT and RoBERTa both loaded	Pass
IT-10	POST /api/analyze	Class, risk level, distress score and word attention returned	Pass
IT-11	Meta-layer reads emotional_stress_scores.csv	500 records, no missing values; used as one of the 11 features	Pass

6.3 Overall Results

Table 6.2 summarises the main result of each component and of the fused system.

Table 6.2: Summary of component and integrated results

Component	Main result
Academic Trend Prediction	Week 17 F1 0.7227, AUC 0.8831; AUC 0.78 at Week 8
Emotional Burnout Detection	Accuracy 82.32%, weighted F1 0.8232, macro F1 0.789
Behavioural Anomaly Detection	OULAD AUC 0.6039 (Version 2), 0.6134 at 0-week lag; SLIIT synthetic AUC 0.7649
Meta-XAI decision layer	Trajectory, priority and workflow implemented; demonstrated on demo data (no outcome data for evaluation yet)
Fusion prototype (Meta-Integration FNN)	F1 0.6836, AUC 0.8576, accuracy 77.23%, precision 0.6065, recall 0.7833 (threshold 0.45); feasibility check only

Permutation analysis attributed about 55% of the fused prediction to the academic signal, 30% to the behavioural signal and 15% to the emotional signal. The academic signal dominates because it is trained directly on course outcomes, while the behavioural and emotional signals help in cases the academic signal misses, such as students who are disengaging or writing distressing text before grades are affected.

These fused results must be read as a feasibility check only. The fusion was also trained before the Version 2 behavioural detector, so it does not yet use the improved curriculum-compliance score. The academic and behavioural predictions use different student identifiers, so the behavioural inputs were simulated from the academic risk, and because no public dataset links student writing to OULAD records, the emotional scores (real outputs of the text model) were not matched to the same students; only the academic-risk inputs were genuinely linked; the 55% academic share is therefore expected, since it is the only input built from real outcomes. The alerts were distributed as about 29% HIGH, 15% MEDIUM and 56% LOW. A HIGH share this large would be too many for advisors to follow up, so the thresholds will need recalibrating once real linked inputs are available (IT-08).

6.4 Stakeholder and User Validation

No formal user study with advisors was conducted within the project timeline. The integrated system and dashboard were demonstrated to the supervisors at the progress presentations, and their feedback led to changes such as clearer per-student explanations on the dashboard. Validation with practising advisors is therefore identified as future work (Section 8.3).

6.5 Achievement of Common Objectives

Table 6.3: Achievement of group objectives

Objective	Status	Justification
G1	Achieved	Risk predicted at four windows; $AUC \geq 0.75$ from Week 8
G2	Achieved	Seven-class classification with phrase-level explanations
G3	Partly achieved	Signal above random up to 8 weeks, but OULAD AUC is modest (0.6039)
G4	Partly achieved	Fusion pipeline works end to end (AUC 0.8576) and Meta-XAI produces prioritised actions, but behavioural and

		emotional inputs are not yet linked per student
G5	Achieved	API and dashboard working; integration tests passed except IT-08

6.6 Integrated Limitations and Threats

- The fusion inputs are not yet linked per student: behavioural inputs were simulated because the academic and behavioural outputs use different student identifiers, and no public dataset links student writing to OULAD records; the fused results therefore show feasibility, not validated detection performance. Exporting the original OULAD identifiers with the academic predictions would allow the behavioural link to be made real.
- The Meta-XAI priority rules and the intervention-opportunity model have not yet been evaluated on real outcome data.
- The fusion thresholds produced about 29% HIGH alerts, which is too many for advisors and needs recalibration with real inputs.
- The text component was trained on social-media posts, not real student writing, and the OULAD-based components come from a UK distance-learning setting, so transfer to Sri Lankan universities is untested.
- The academic signal dominates the fusion, so errors in that component have the largest effect on the final alert.
- The behavioural signal on OULAD is modest (AUC 0.6039); the VAE reconstruction error alone is below chance (AUC 0.4261) because chronically disengaged students look normal to it, so most of the signal comes from the curriculum-compliance term. The higher SLIIT result (AUC 0.7649) comes from synthetic data and is not real-world evidence.
- The system has not been evaluated with real advisors, so its effect on intervention decisions is unknown.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

For universities, the system offers earlier and explained identification of at-risk students, which can reduce withdrawals and the related loss of tuition income. For advisors, it replaces several separate indicators with one prioritised list in which every alert has a reason. For students, it offers the chance of support before problems become severe. For society, it supports retention and student well-being.

7.2 Market Alternatives and Differentiation

Table 7.1: Comparison with existing solutions

Feature	Moodle Analytics [7]	OU Analyse [6]	Commercial platforms (e.g. EAB Navigate, Civitas Learning)	R26-IT-059 system
Signals used	LMS engagement	Demographic and VLE data	Mainly academic and enrolment data	Academic, emotional text and LMS behaviour, with prioritised advisor actions
Timing	Chosen time splits	Weekly	Varies by vendor	Weekly, with four prediction windows
Per-student explanation	No	Limited	Limited / not published	Yes, from every signal
Emotional well-being	No	No	No	Yes, with escalation
Cost / access	Free, open source	Internal to the Open University	Paid licence	Planned low-cost plugin

The main differentiation is the combination of several signals with per-student explanations from each of them, including emotional well-being, which existing dropout tools do not cover.

7.3 Business Model and Feasibility

A freemium model is proposed: a free tier providing the academic risk list for one course, and a paid tier per enrolled student per year that adds the emotional and behavioural signals, fusion and explanations. Likely channels are direct sales to universities and an LMS plugin. Technical feasibility is supported by the working prototype; operational feasibility is supported by batch processing, which keeps hosting costs low. Exact pricing and break-even depend on institutional pilots.

7.4 Adoption Intellectual Property and Compliance

The main adoption barriers are trust in AI decisions, privacy concerns about analysing student text and the need for staff training. These are addressed by explanations, human review and data minimisation. In Sri Lanka, the Personal Data Protection Act No. 9 of 2022 became fully enforceable on 18 March 2025 [8], so deployment would require a lawful basis for processing, transparency to students, data minimisation and security safeguards; analysing student text would additionally require explicit consent. The code is the team's own work, and the datasets are used under their published licences [3], [4]. Any real deployment would also need ethics approval and informed consent before collecting student text; the emotional output must never be used as a diagnosis, and Suicidal-category outputs must always be routed directly to qualified mental-health professionals.

7.5 Sustainable Development Goal Contribution

The system can plausibly contribute to SDG 4 (target 4.3) by supporting retention and to SDG 3 (target 3.4) by flagging emotional distress early. Possible negative effects include stigmatising flagged students, false alarms and biased predictions for under-represented groups. To sustain the benefits, alerts should remain confidential, every alert should be reviewed by a person, and fairness across student groups should be audited regularly.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

The R26-IT-059 project shows that burnout and dropout risk can be detected earlier and explained better by combining several signals than by relying on grades alone. The academic component makes alerts reliable from Week 8, the emotional and behavioural components detect students the academic signal can miss, and the meta-integration layer shows that all signals can be combined into one prioritised alert with reasons from each signal. The fusion prototype (AUC 0.8576) confirms that the pipeline works end to end, but because its behavioural inputs were simulated and its emotional scores were not linked to the same students, the common problem is addressed in principle only; validation on consented, linked student data and with advisors is still required.

8.2 Principal Integrated Contribution

Beyond the four components, the team produced an explainable multi-modal early warning system with a shared data contract, a late-fusion layer that measures how much each signal contributes, and a single advisor dashboard. The team also showed that the signals are complementary, which supports combining them rather than improving any one model alone.

8.3 Future Directions

- Collect consented, linked text and LMS data from real students so that the fusion can be trained and evaluated on real inputs, then recalibrate the alert thresholds.
- Validate the system with real advisors and, with consent, with SLIIT student data.
- Collect consented student text to reduce the domain shift of the emotional component.
- Deploy the system on a hosted server with scheduled weekly runs, monitoring and role-based access.
- Audit fairness across student groups before any operational use.

REFERENCES

- [1] E. Yeskuatov, L. K. Foo, and S.-L. Chua, "Detecting burnout among undergraduate computing students with supervised machine learning," *Healthcare*, vol. 13, p. 3182, 2025, doi: 10.3390/healthcare13233182.
- [2] B. Alnasyan, M. Basher, and M. Alassafi, "The power of deep learning techniques for predicting student performance in virtual learning environments: A systematic literature review," *Comput. Educ.: Artif. Intell.*, vol. 6, p. 100231, 2024.
- [3] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," *Sci. Data*, vol. 4, p. 170171, 2017.
- [4] S. Sarkar, "Sentiment analysis for mental health," Kaggle Dataset, 2024. [Online]. Available: <https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>
- [5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.
- [6] C. Herodotou, M. Hlosta, A. Boroowa, B. Rienties, Z. Zdrahal, and C. Mangafa, "Empowering online teachers through predictive learning analytics," *Br. J. Educ. Technol.*, vol. 50, no. 6, pp. 3064–3079, 2019, doi: 10.1111/bjet.12853.
- [7] Moodle, "Students at risk of dropping out," Moodle Docs. [Online]. Available: https://docs.moodle.org/en/Students_at_risk_of_dropping_out
- [8] Data Protection Authority of Sri Lanka, "Personal Data Protection Act dates of operation announced." [Online]. Available: <https://dpa.gov.lk/est.php>
- [9] X. Xu et al., "GLOBEM: Cross-dataset generalization of longitudinal human behavior modeling," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 6, no. 4, 2023.
- [10] M. Nagy and R. Molontay, "Interpretable dropout prediction: Towards XAI-based personalized intervention," *Int. J. Artif. Intell. Educ.*, vol. 34, pp. 274–300, 2024.

APPENDICES

Appendix A Integrated Interface Specification

This appendix specifies the REST interface of the integrated system. All responses are JSON; student identifiers are anonymised.

Table A.1: REST API endpoints

Endpoint	Purpose	Errors
GET /health	Service status and number of loaded predictions	500 if predictions not loaded
GET /students?alert=HIGH MEDIUM LOW	All students, optionally filtered by alert level	–
GET /students/<idx>	One student by index	404 if out of range
GET /search?id=<student_id>	One student by identifier	400 if ID missing; 404 if not found
GET /summary	Alert-level counts and model metrics	–

Example academic record: {"student_id": "OULAD_TEST_0000", "academic_risk": 0.2561, "week4_risk": 0.5022, "week8_risk": 0.1955, "week12_risk": 0.3702, "week17_risk": 0.2561, "alert_level": "LOW", "top_feature_1": "F1_login_count", "top_feature_1_pct": 17.7}

Appendix B Team Contribution Declaration

The contributions of each member are summarised in Table 4.3. All four members confirm that this record is accurate and agreed.

Table B.1: Team contribution declaration

Member	Student ID	Main individual contribution	Shared contribution	Signature
Mahavithana S. M.	IT22916426	Academic trend prediction, API	Joint paper, testing, report	
Induwara K. P. Y.	IT22196392	Emotional burnout detection	Joint paper, testing, report	
Karunaratne D. C.	IT22215710	Behavioural anomaly	Joint paper, testing, report	

		detection		
Jayawickrama G. T.	IT22253194	Meta-integration, Meta-XAI decision layer, dashboard	Joint paper, testing, report	

AI use disclosure: generative AI (Claude, Anthropic) was used to help structure and draft this report from the team's own papers, theses and results, and to check language and references. All results come from the team's own experiments, and all content was reviewed and verified by the team.